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# Experiment Configuration
Element Budget: 25
Gamma_C: 50.0
Max Episode Steps: 200
Entropy Coefficient: 0.01
Learning Rate: 0.0003
```

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# Adaptive Mesh Refinement RL Training Summary
Date: 2025-03-24 15:34
Total Steps: 50000
Episodes Completed: 1255
```

Termination Statistics

- Budget exceeded (pre-action): 57 (4.5%)
- Budget exceeded (post-adapt): 41 (3.3%)
- Error: Mesh quality issues: Element size ratio too large: 384.00: 56 (4.5%)
- Maximum consecutive no-actions reached: 1100 (87.6%)
- Maximum episode steps reached: 1 (0.1%)

Action Distribution

- Coarsen: 2479 (5.0%)
- No Change: 37040 (74.1%)
- Refine: 10481 (21.0%)

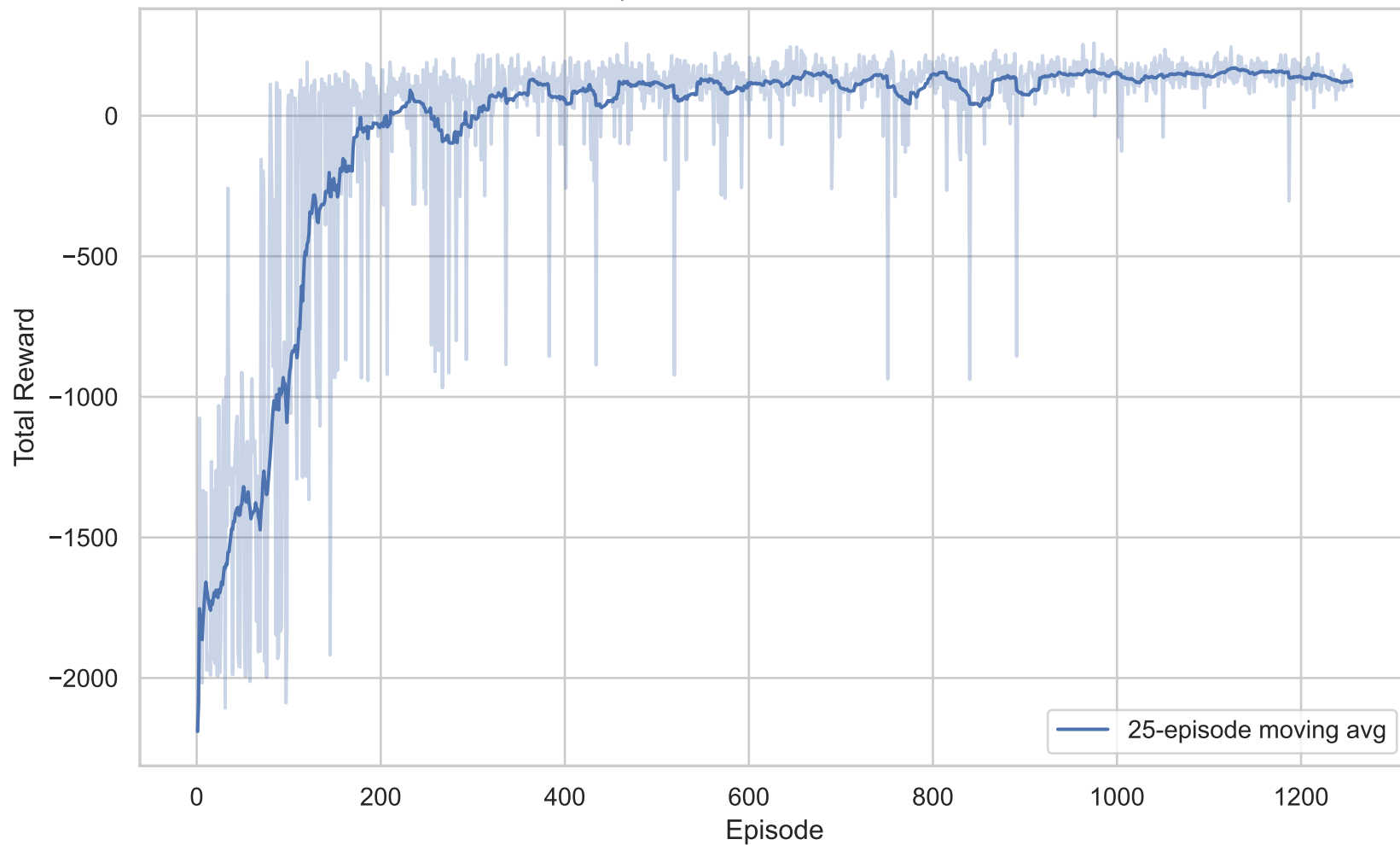
Reward Statistics

- Mean Reward: -32.08
- Min Reward: -2190.20
- Max Reward: 258.15
- Mean Reward (last 251 episodes): 143.19

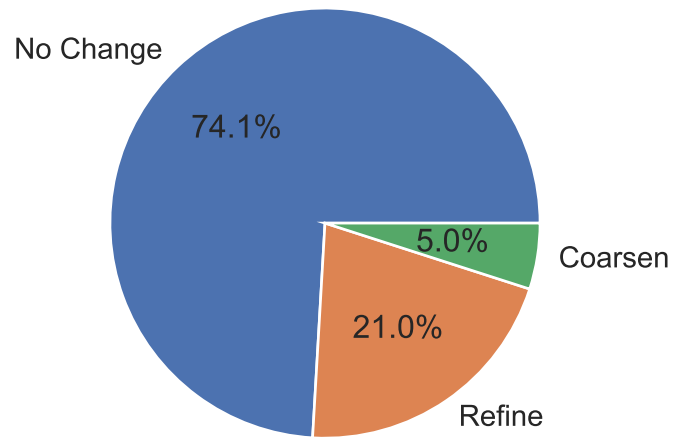
Resource Usage

- Mean Resource Usage: 0.46
- Max Resource Usage: 1.00
- Episodes ending with budget exceeded: 0 (0.0%)

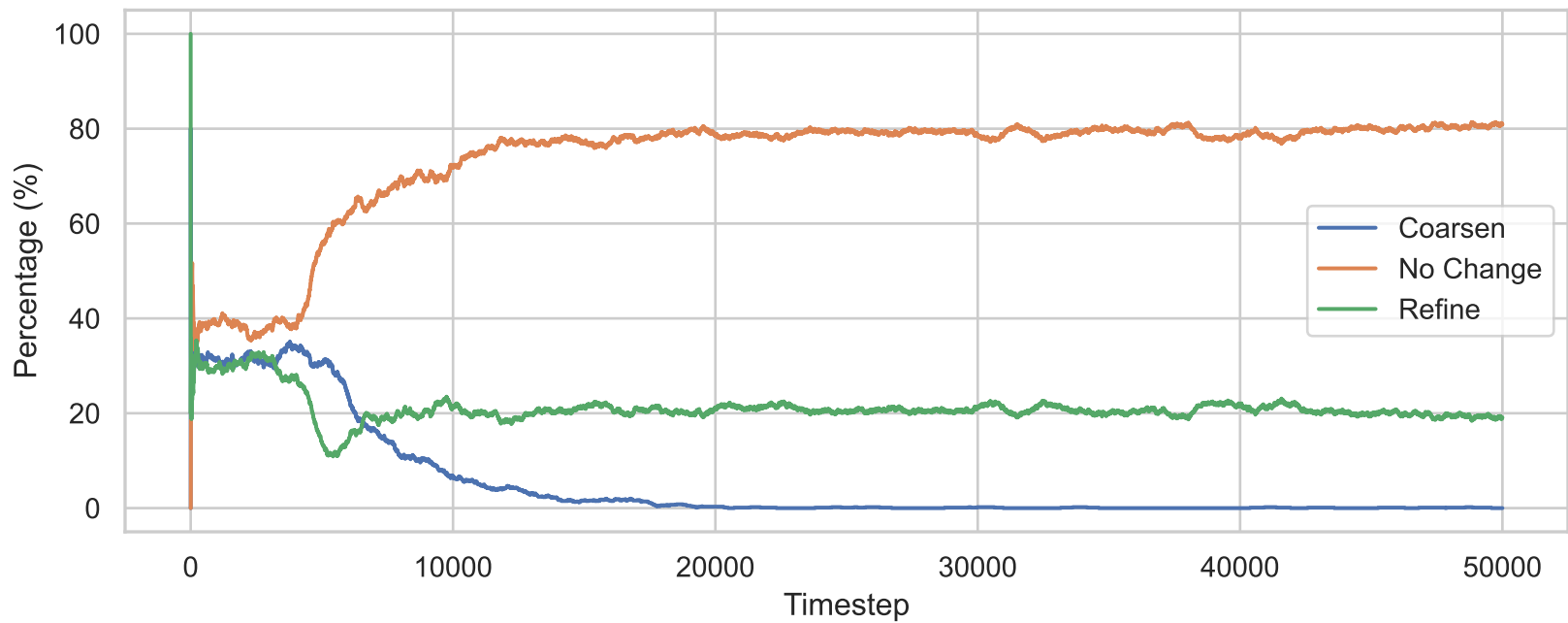
Episode Rewards Over Time

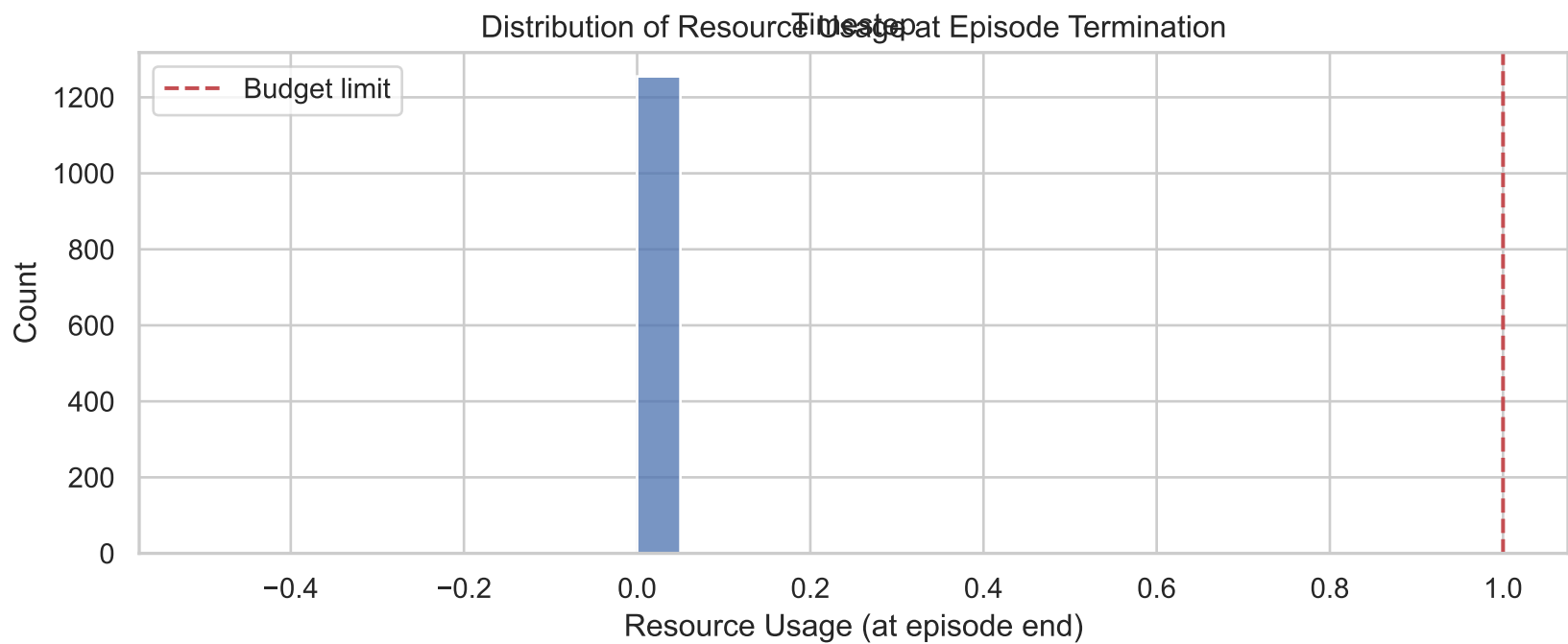
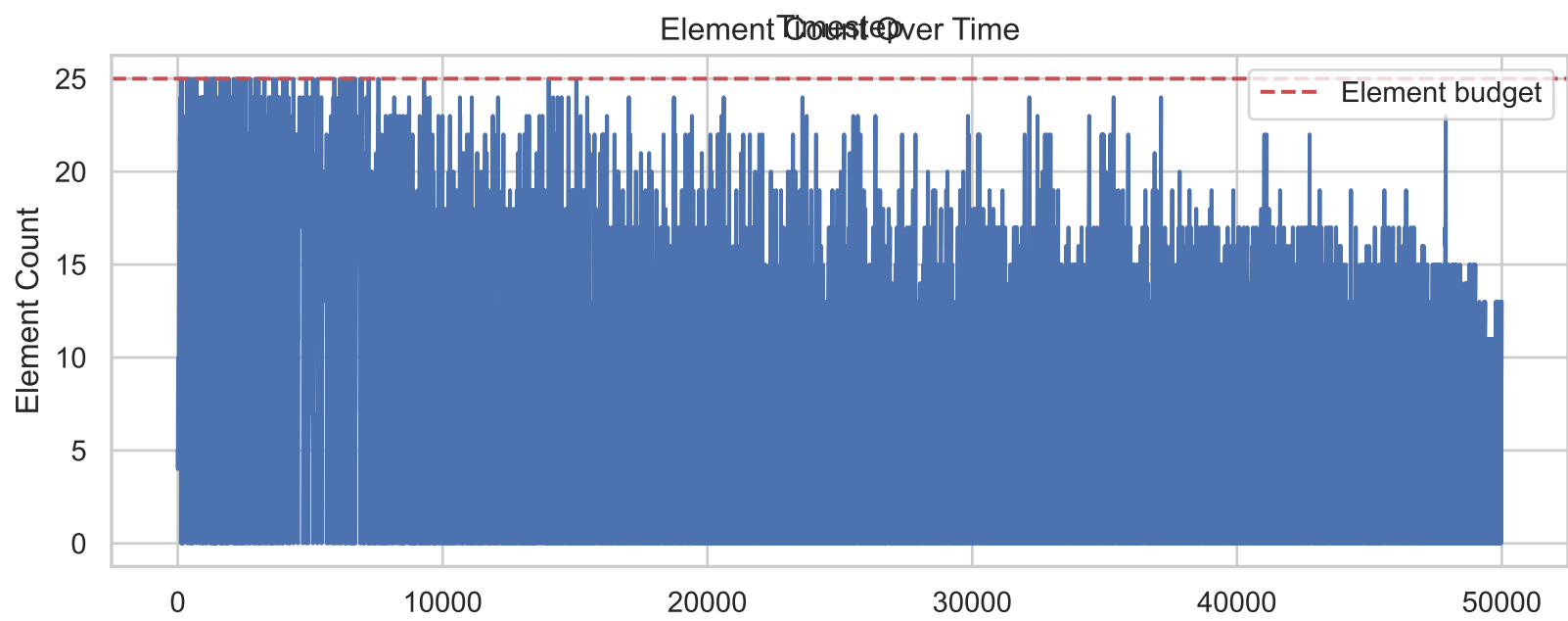
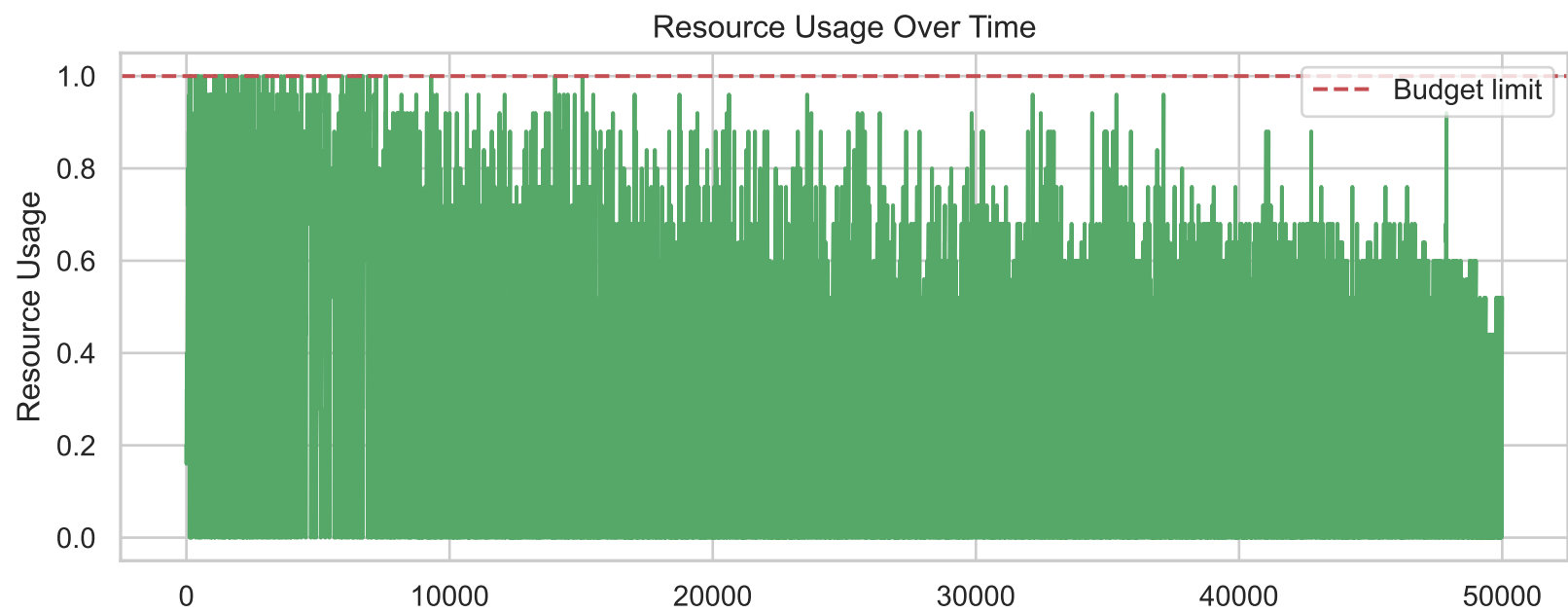


Overall Action Distribution

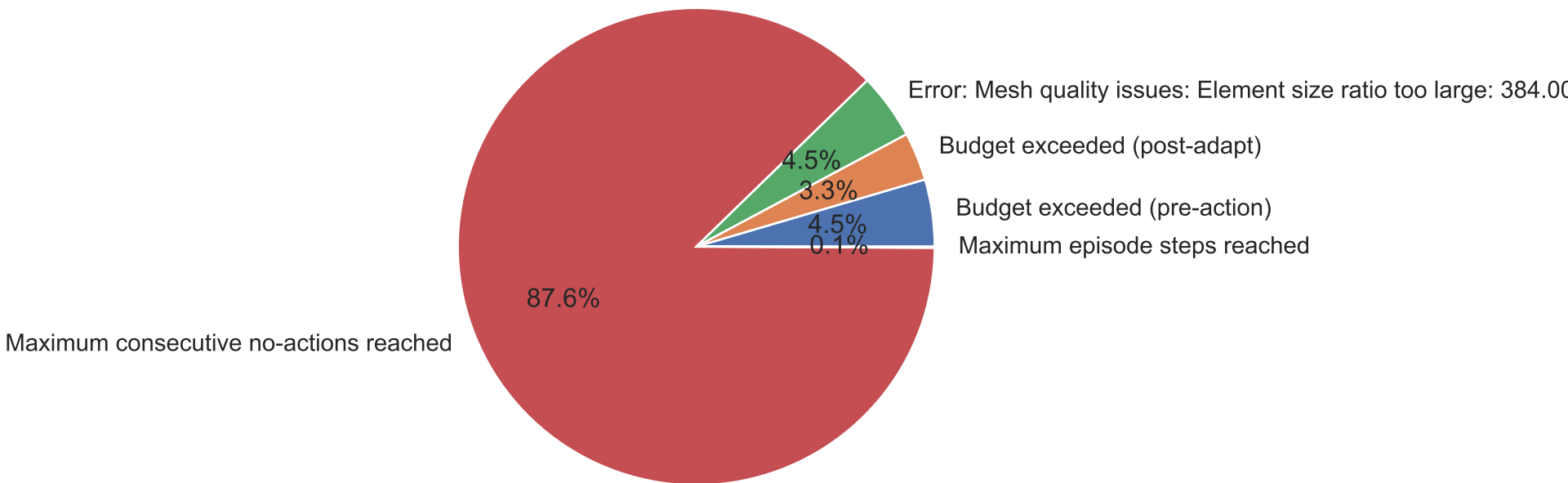


Action Distribution Evolution

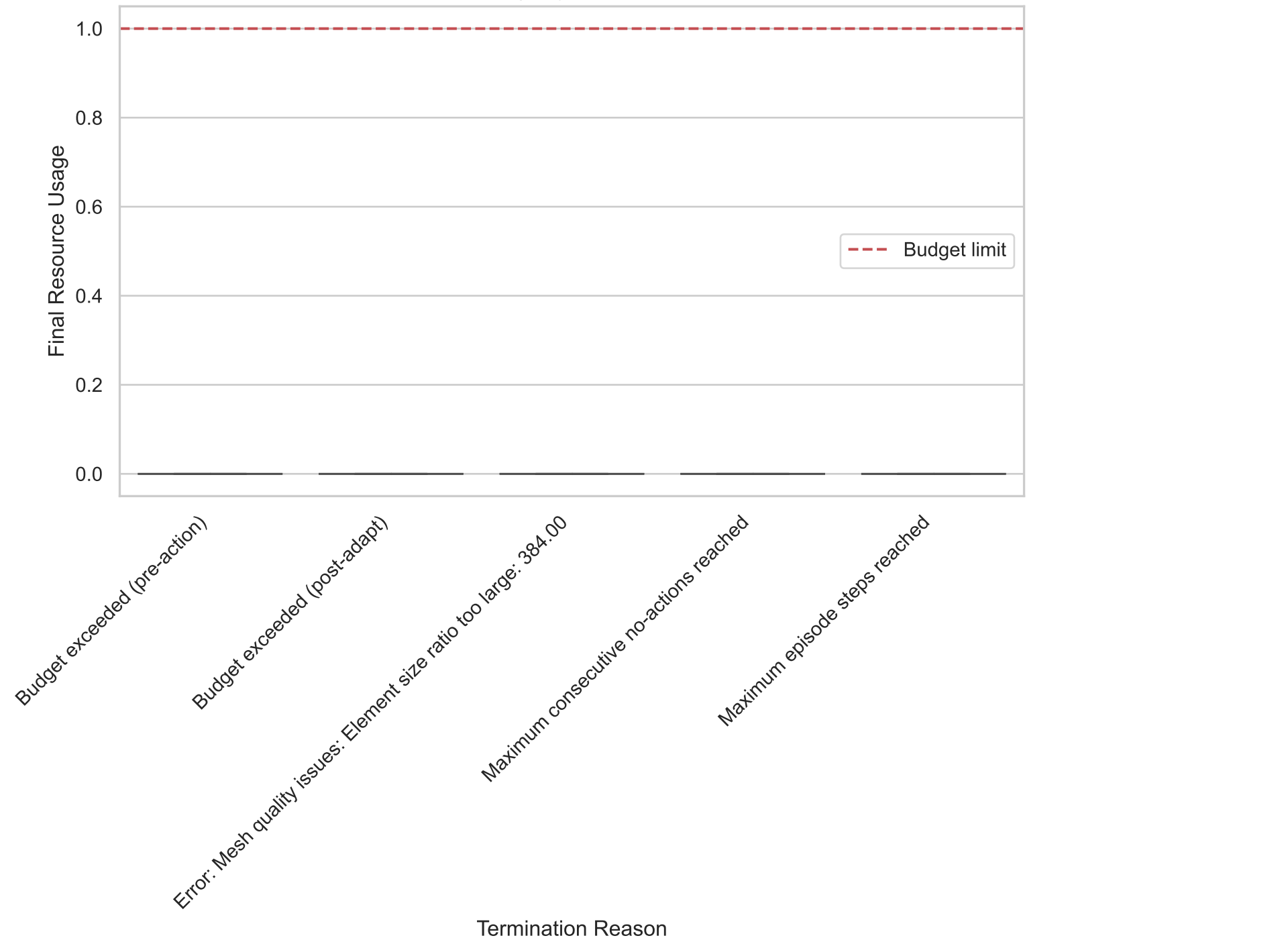




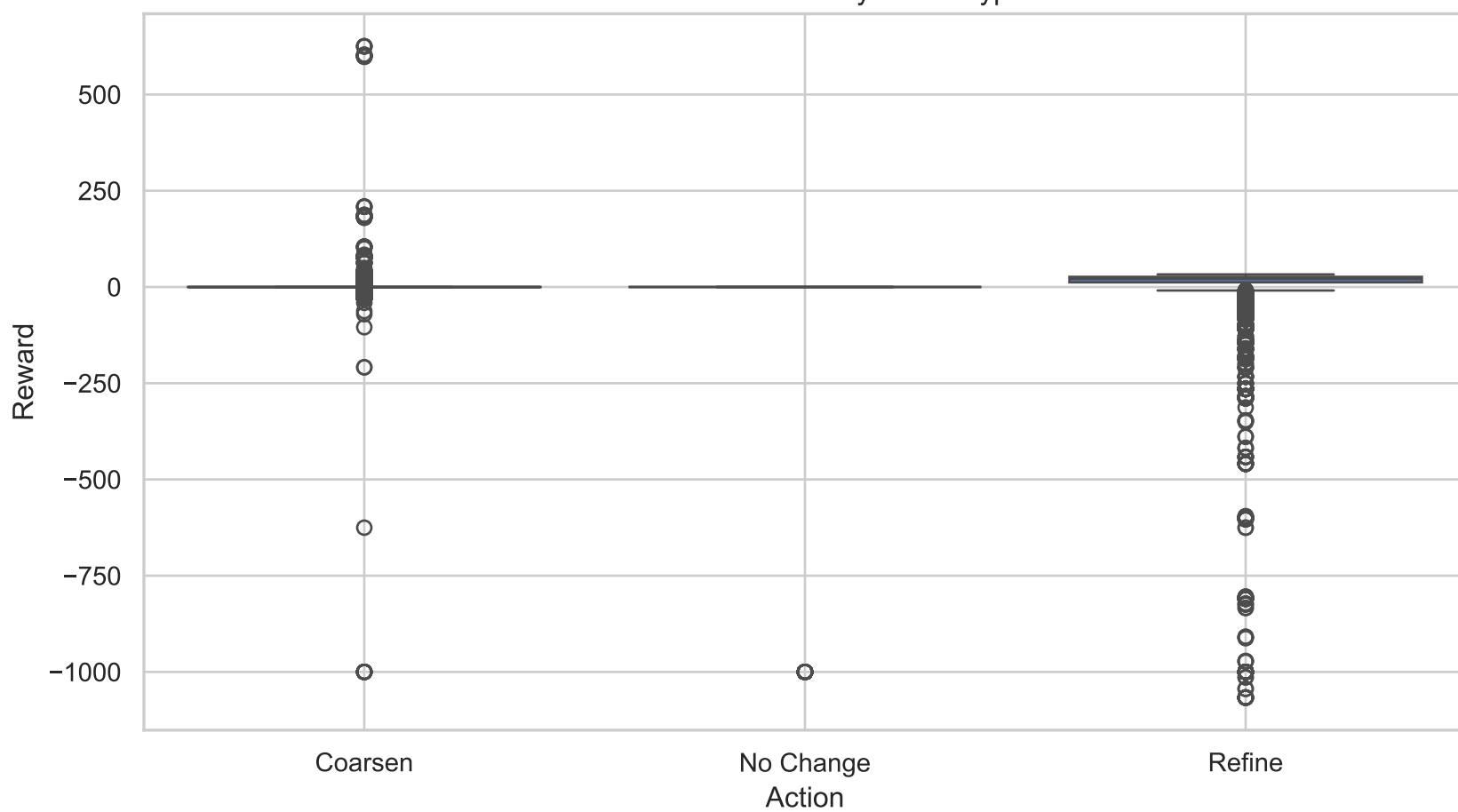
Episode Termination Reasons



Resource Usage by Termination Reason

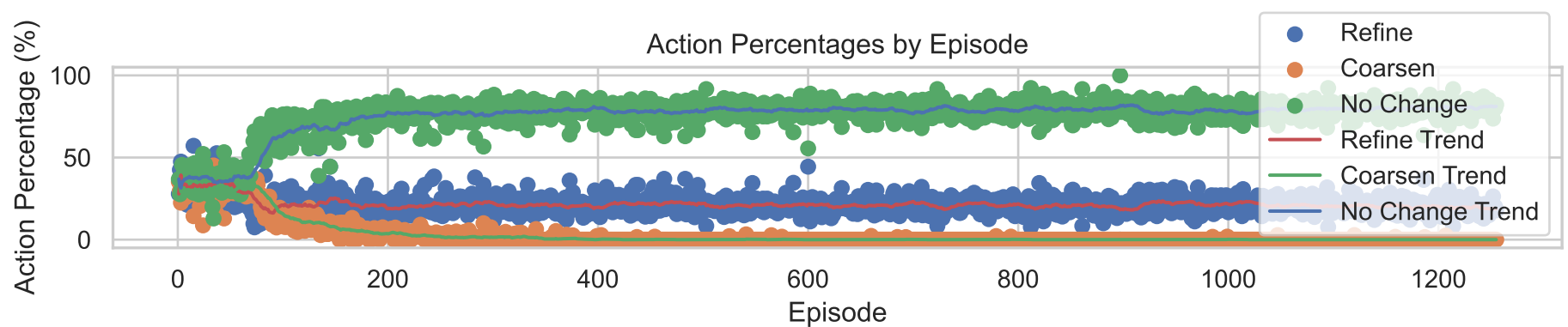
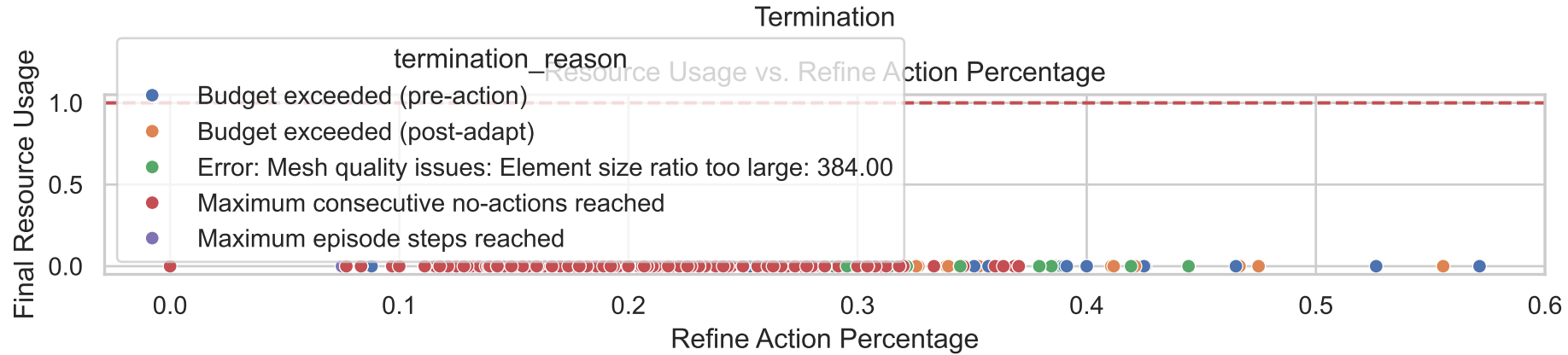
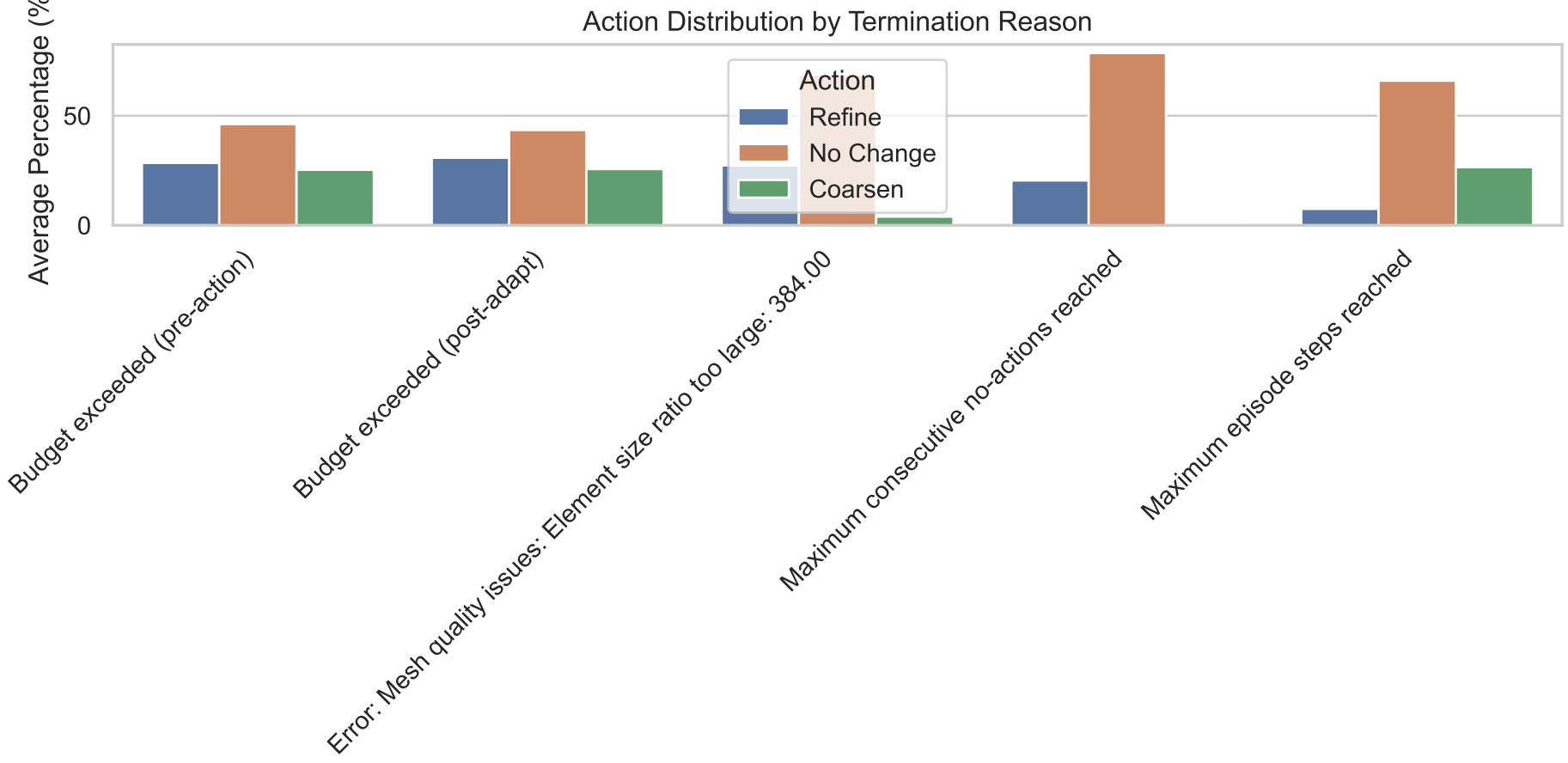


Reward Distribution by Action Type

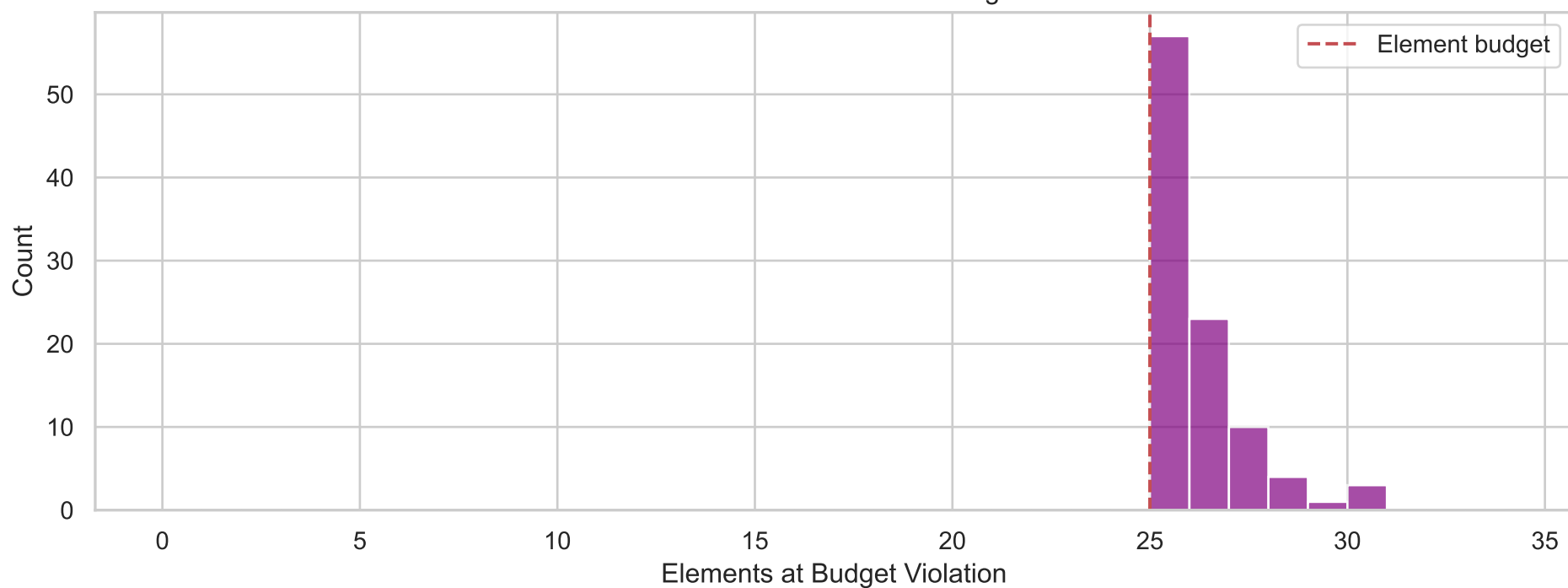


Episode Length vs Reward

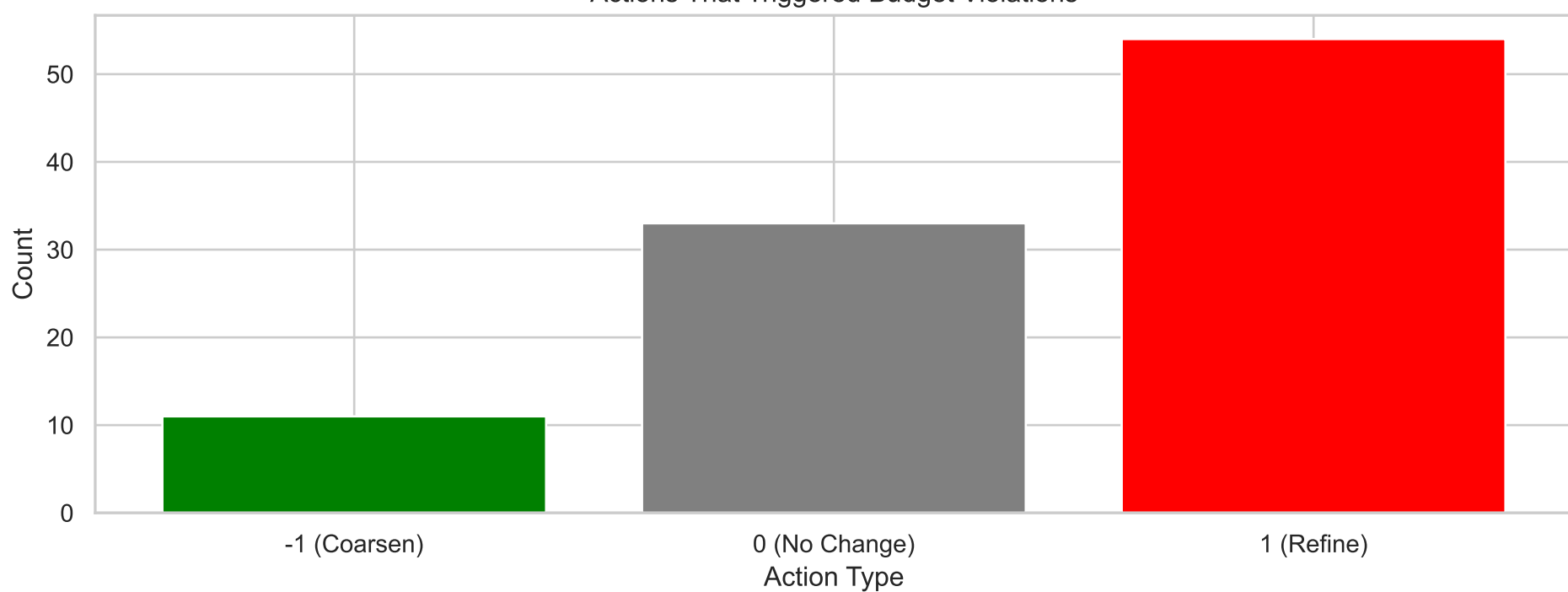




Distribution of Element Counts at Budget Violation



Actions That Triggered Budget Violations



Budget Usage Percentage at Violation

